數位電路實驗結報一

–6-bit 數位加法器–

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1 實驗理論與原理

1.1 半加器與全加器 (Half Adder & Full Adder)

加法器為數位邏輯電路中執行二進位加法運算之核心組件。最基本的**半加器**負責兩個單一位元 A 與 B 的加法，其輸出包含總和 (S) 與進位 (C_{out})，布林邏輯表示式為：

$$\mathrm{S} = \mathrm{A} \oplus \mathrm{B} \tag{1}$$

$$\mathrm{C_{out}} = \mathrm{A} \cdot \mathrm{B} \tag{2}$$

為了處理多位元的加法，必須考慮來自較低位元的進位。因此引入了**全加器**，它包含三個輸入端：兩個加數 (A, B) 以及一個進位輸入 (C_{in})。全加器的輸出邏輯表示式為：

$$\mathrm{S} = \mathrm{A} \oplus \mathrm{B} \oplus \mathrm{C_{in}} \tag{3}$$

$$\mathrm{C_{out}} = (\mathrm{A} \cdot \mathrm{B}) + (\mathrm{C_{in}} \cdot (\mathrm{A} \oplus \mathrm{B})) \tag{4}$$

1.2 邏輯閘實現與笛摩根定理

在實體電路佈線中，我們使用 IC 7486 (XOR 閘) 與 IC 7400 (NAND 閘) 來建構上述邏輯。總和 S 可直接透過兩個 XOR 閘串接完成。

對於進位 C_{out} 的部分，原本的設計需要 AND 閘與 OR 閘。為了僅使用 NAND 閘 (IC 7400) 來完成進位電路，我們可套用**笛摩根定理 (De Morgan's Laws)** 將原本的 AND-OR 形式轉換為 NAND-NAND 形式：

$$C_{out} = \overline{\overline{(A \cdot B)} \cdot \overline{(C_{in} \cdot (A \oplus B))}} \tag{5}$$

透過此轉換，即可大幅簡化所需的 IC 種類，完全依靠 7486 與 7400 實現 1-bit 全加器。

1.3 漣波進位加法器 (Ripple Carry Adder)

要實現多位元（如本實驗之 6-bit）的加法運算，可將多個全加器串聯組成「漣波進位加法器」。其核心原理是將第 i 個位元的進位輸出 ($C_{out,i}$) 直接連接至第 $i+1$ 個位元的進位輸入 ($C_{in,i+1}$)。

本次實驗中，我們將兩個現成的 3-bit 加法器模組進行擴充，把低位元 (Lower 3-bits) 加法器的最高位進位輸出，連接至高位元 (Upper 3-bits) 加法器的最低位進位輸入 (C_{in})，藉由進位訊號的傳遞 (Carry Propagation)，成功構建出完整的 6-bit 數位加法器。

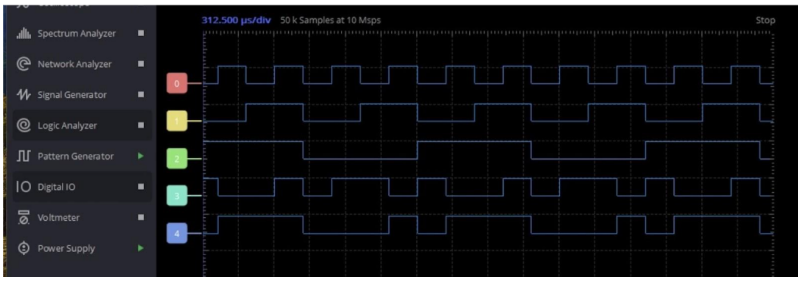
2 實驗架設與步驟

2.1 1-bit 數位加法器實作 (實驗一)

1. **硬體接線：**

   - 準備 IC 7486 (XOR) 與 IC 7400 (NAND)，將其固定於麵包板上。分別將兩顆 IC 的第 14 腳位連接至 ADALM2000 提供的 +5V 電源 (VCC)，第 7 腳位連接至接地 (GND)。

- 依據理論化簡後之電路圖進行佈線：利用 7486 負責處理總和 (S) 與部分進位邏輯，並將其輸出結果接入 7400 進行 NAND 邏輯運算，以產生最終的進位輸出 (C_{out})。
- 將 ADALM2000 的數位腳位 (DIO) 連接至電路對應節點：DIO 0 接至輸入 A、DIO 1 接至輸入 B、DIO 2 接至輸入 C_{in}；量測端的部分，將 DIO 3 接至總和輸出 S、DIO 4 接至進位輸出 C_{out}。



2. **軟體設定 (Scopy)：**
   - **Pattern Generator (訊號產生器)：**開啟 Scopy 軟體，將 DIO 0、1、2 設定為 Binary Counter 模式，並設定為三個等比頻率（例如 A 設為 4 kHz、B 設為 2 kHz、C_{in} 設為 1 kHz）。此設定可產生所有 8 種 0 與 1 的輸入排列組合。
   - **Logic Analyzer (邏輯分析儀)：**開啟 DIO 0 至 DIO 4 的波形顯示。將取樣率 (Sample Rate) 設定為 5 Msps，取樣數量 (Nr of samples) 設定為 20k Samples。
   - 擷取波形，觀察輸出腳位與輸入腳位之電位變化，並比對全加器真值表以驗證硬體邏輯。

2.2 多位元加法器串聯測試 (實驗二與延伸)

1. **3-bit 加法器模組測試：**取得助教事先製作的 3-bit 加法器電路板，透過板上的指撥開關給定不同的輸入組合，並觀察 LED 燈亮的狀態來驗證 3-bit 的加法結果（例如測試 $3 + 5 = 8$，即二進位之 $011_2 + 101_2 = 1000_2$）。
2. **6-bit 加法器擴充：**將兩塊 3-bit 加法器模組進行串聯。把處理低位元 (Lower 3-bits) 模組的進位輸出端 (C_{out})，利用跳線連接至處理高位元 (Upper 3-bits) 模組的進位輸入端 (C_{in})。
3. 透過 ADALM 再次進行輸入測試，確認進位訊號能順利跨模組傳遞，完成並驗證完整的 6-bit 加法運算邏輯。

3 實驗結果與分析

4 問提與討論

4.1 請比較 AC 磁化率跟 DC 磁化率的差別

AC magnetic susceptibility and DC magnetic susceptibility are two ways to measure how a material responds to an external magnetic field, but they operate under different conditions and reveal different magnetic behaviors:

- Frequency of the Magnetic Field:

  DC magnetic susceptibility is measured in the presence of a steady, unchanging magnetic field (direct current, or DC field). It reflects the material's magnetic response to a constant magnetic field.

  AC magnetic susceptibility is measured when an alternating magnetic field (alternating current, or AC field) is applied. The field changes direction periodically, allowing researchers to study the dynamic response of the material over time.

- Measurement of Different Magnetic Properties:

  DC susceptibility measures the equilibrium response of a material, showing how much it is magnetized when exposed to a constant magnetic field. It gives insights into the material' s overall magnetic moment and bulk properties.

  AC susceptibility gives more information about the dynamic processes in the material, such as the motion of domain walls, relaxation times (Néel or Brownian relaxation in nanoparticles), and other frequency-dependent magnetic behaviors. AC measurements are particularly useful in studying magnetic systems with time-dependent properties.

- Applications:

  DC measurements are simpler and are often used for basic magnetic characterizations, like identifying whether a material is paramagnetic, ferromagnetic, or diamagnetic.

  AC measurements are more sophisticated and are commonly used in research areas that involve magnetic relaxation (e.g., magnetic nanoparticles), studying the frequency response of materials, and detecting magnetic phase transitions.

In summary, DC magnetic susceptibility is a static measure of how a material reacts to a constant magnetic field, while AC magnetic susceptibility provides insights into the material's response to a time-varying field, offering more detailed information about dynamic magnetic processes.

5 總結

In this experiment, we measure the susceptibilities of Fe_3O_4 ferrofluid with different concentration and temperature. Then, we use the susceptibility to derive the concentration of the unknown sample. The concentrations of the unknown sample in different temperature are consistent, approximately **4.8 mg/ml**. Additionally, we investigate the susceptibility of ink and find its susceptibility is weak.

In addition, we measured the samples under different frequencies. Though susceptibility data get severely distorted at lower frequency regions, the region over 500 Hz fits relatively well with equation of magnetic particles in fluid. The time constant for iced ferrofluid is way larger than that under room temperature, indicating relaxation happens slower at lower temperatures. Finally, when doing the YIG sample, the curves cannot match the equation of magnetic particles in fluid. This exemplifies that YIG is not isotropic, and its relaxation mechanism is different from ferrofluids.